From Approval to Advance

A Data-Driven Forecast of Mortgage Completions and Two-Month Cashflow

Chin-Ju (Wendy) Lin

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Executive Summary

This project develops models to predict mortgage completion probability, completion duration, and expected two-month cashflows. The modelling approach prioritises identifying likely non-completion cases, as these create cashflow and balance-sheet planning risk. The final framework combines completion probability and time-to-completion predictions to produce a probabilistic cashflow forecast.

1. Business Context and Objectives

This project addresses two predictive tasks: completion probability prediction and time-to-completion prediction. The first task estimates the likelihood that an approved loan application will proceed to final completion. The second task estimates the time required to complete the pipeline for applications that are expected to reach completion.

2. Data Exploration and Preparation

2.1 Information Framework

To structure the analysis, the variables are grouped into five strategic pillars: Loan Metrics, Application Source, Application Demographics, Property Attributes, and Time Information. This framework provides a clearer basis for understanding how different types of information may influence the two target outcomes.

2.2 Target Definition

Two predictive tasks are defined for this study. For the completion probability model, an application is classified as completed if an advance date is recorded. For the time delay model, the target variable is defined as the number of days between the approval date and the advance date.

2.3 Data Integrity Checks

Applications from the most recent three months are excluded. These cases show unusually low or even zero completion rates, because many are likely still in progress and do not yet have sufficient time to complete the pipeline. Excluding these recent cases reduces distortion in the target variable and ensures that

observed completion outcomes better reflect actual performance rather than incomplete process maturity.

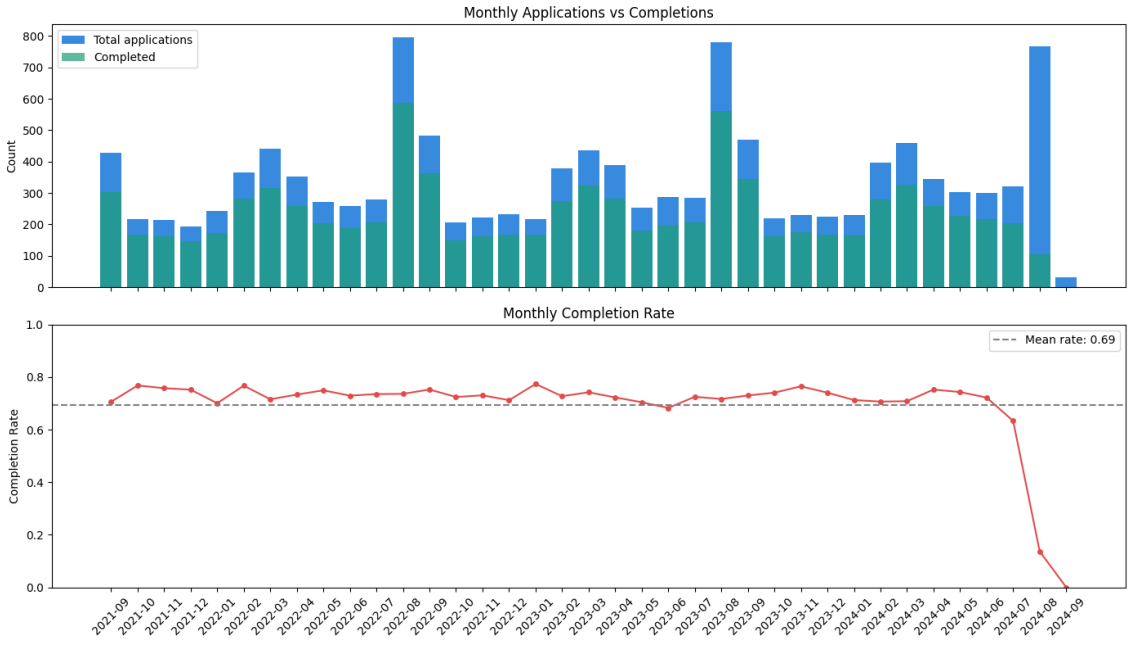


Figure 1: Monthly Applications, Completions, and Completion Rate

In addition, negative completion durations are removed, as these values are not operationally valid and would reduce the quality of model training and evaluation.

2.4 Data Splitting and Preprocessing

A chronological split is used, with applications before 1 January 2024 used for training and later observations are reserved for testing. This better reflects a real forecasting setting and reduces the risk of data leakage. All imputation and derived features that depended on historical outcomes are learned from the training data only and then applied to the test set. Missing property age is imputed using the training median, while missing second-applicant income is set to zero where no second applicant was present.

2.5 Feature Engineering

Additional features are created to better represent the financial, property, and time-related characteristics of each application. These engineered variables are grouped into leverage, affordability, interaction, timing, and transformation features.

LTV is calculated as loan amount divided by property valuation, and a high-LTV flag is created to identify applications with an LTV above 0.85. To capture affordability, total applicant income is calculated by combining the income of the first and second applicants, and an income multiplier is created by dividing the loan amount by total income. An interaction variable between LTV and the income multiplier is also added to represent cases where both leverage and borrowing burden are high.

Time-related features are created from the approval date, including approval month and approval quarter. In addition, monthly average completion delay is calculated from the training data and then mapped to both the training and test sets to avoid data leakage. Finally, log-transformed versions of loan amount, property valuation, and total income are added to reduce the influence of extremely large values. A new-build indicator is also created based on whether property age is equal to zero.

Overall, these engineered features are designed to provide a more meaningful representation of each application before model development.

3. Critique of the Analyst's Approach and Key Improvements

3.1 Issues in the probability model

I modify the analyst's model mainly to improve the quality and relevance of the input features. The original model uses only country, business source, and loan amount, which may not fully capture the key drivers of completion risk. Therefore, I expand the feature set to include more meaningful applicant, affordability, and property-related variables. This allows the model to better reflect the factors that are likely to influence whether an application is completed.

I also change the way categorical variables and missing values are handled. Instead of treating category codes as continuous numeric values, categorical variables are processed more appropriately to avoid imposing artificial ordering. Missing values are not simply replaced with zero; instead, numerical variables are imputed using median values, and categorical variables are imputed using the mode or treated as a separate category where necessary. These modifications make the model more reliable, interpretable, and better aligned with the structure of the data.

3.2 Issues in the time delay model

I modify the analyst's time delay model because the original model includes uncompleted applications, even though time to completion is only observed for completed cases. To address this, I restrict the model to completed applications only, ensuring that the target variable represents a valid realised duration.

I also avoid filling missing time-to-complete values with zero, because this would incorrectly treat uncompleted applications as if they completed immediately. Excluding these cases from the delay model prevents distortion of the target variable and makes the regression results more meaningful. Overall, these modifications make the model more consistent with the business problem and improve the reliability of the predicted completion duration.

3.3 Process improvements

Because the business objectives are to predict outcomes for newly approved applications, model validation should be based on a temporal split rather than a random split. Earlier approvals should be used for model development, while later approvals are reserved for out-of-sample testing. This approach more closely reflects how the model would be used in practice and reduces the risk of overstating predictive performance.

In addition, using a matrix plot can help clarify the relationships between variables and support more informed feature selection. This improves analytical efficiency and helps identify variables that are more likely to contribute meaningful predictive value.

Further improvement can also be achieved through feature engineering. Creating derived variables from the existing data, such as affordability-related ratios, can provide a more realistic representation of the underlying business situation and improve model robustness.

4. Completion Probability Problem

The value of completion prediction lies in helping the business prepare for applications that may fail to complete. If some approved applications do not proceed through the full pipeline, the company should be able to anticipate this and put in place a contingency plan to manage the resulting cashflow risk. For this reason, the modelling objective is not defined solely in terms of maximising overall accuracy. Greater emphasis is placed on improving the identification of likely non-completion cases, even if this requires accepting a modest trade-off in headline

accuracy. This principle is applied consistently across the completion prediction models, including the choice of classification threshold.

4.1 Improved Linear Model for Completion Probability

4.1.1 GLM Lasso Model Design and Interpretation

A logistic regression model with Lasso regularisation is developed for completion classification. The model uses applicant, loan, affordability, property, and approval timing variables, including both original and engineered features. This provides a broader and more business-relevant input structure than the original model while retaining interpretability.

Lasso regularisation is used to control model complexity by shrinking weaker predictors towards zero. In addition, the classification threshold is adjusted to 0.55 to reduce overly optimistic completion predictions and improve the identification of likely non-completion cases.

The results suggest that the probability of completion is mainly driven by property-related attributes. Among these factors, property age is the most influential predictor, indicating that the condition or lifecycle stage of the property plays an important role in completion outcomes. In addition, the model also highlights the distinction between new and existing properties, suggesting that whether a property is newly built or not has a meaningful impact on completion probability.

4.2 Non-linear Model for Completion Probability

4.2.1 Random Forest and Results

Random forest is chosen because it can capture non-linear patterns and interactions between variables more effectively than a linear model. It does not require a strict assumption about the form of the relationship between predictors and the target.

A random forest model is used as a non-linear benchmark because it can capture interactions and non-linear relationships without requiring a fixed functional form. To maintain comparability, it is built using a focused set of business-relevant predictors and evaluated using the same business-oriented thresholding principle.

This model achieves a ROC AUC of 0.7878, indicating good overall predictive ability with a slightly higher recall for uncompleted applications (0.71) than for completed applications (0.67).

4.2.2 Model Comparison: GLM Lasso Model vs Random Forest

For this problem, the two models show very similar overall discriminative performance. However, the GLM Lasso model is better at identifying applications that remain uncompleted, while the random forest model performs slightly better in identifying completed applications. Although the random forest model achieves marginally higher accuracy, the difference is small.

Model	Accuracy	ROC AUC	Recall for Uncompleted Applications	Recall for Completed Applications
GLM Lasso	0.6489	0.7905	0.80	0.59
Random Forest	0.6810	0.7878	0.71	0.67

Table 1: Comparison of Completion Classification Models

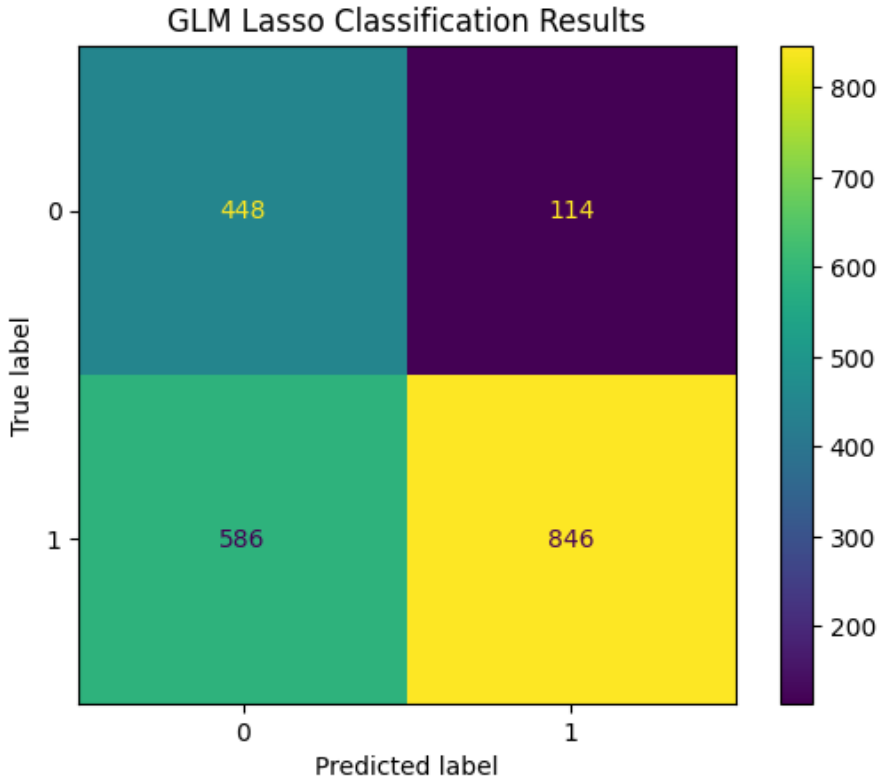


Figure 2: GLM Lasso Classification Results

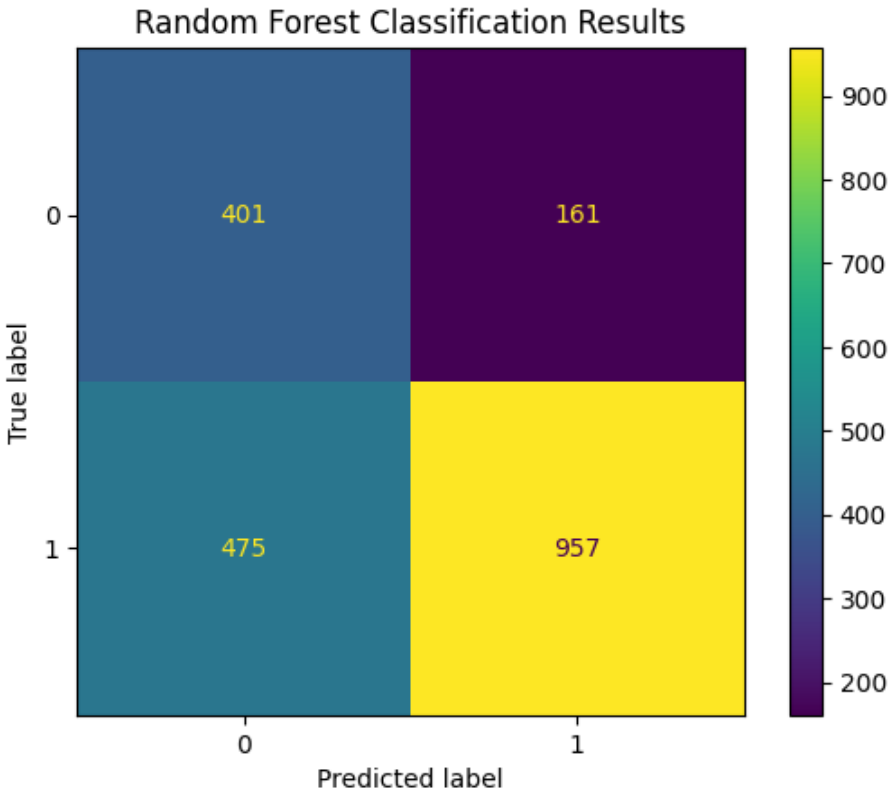


Figure 3: Random Forest Classification Results

4.2.3 Recommendation and Interpretation

The similarity in performance suggests that the completion prediction problem may not contain substantial non-linear structure, meaning that a simpler linear model is already able to capture the main patterns in the data. However, the GLM Lasso model is preferred because it is easier to interpret, less computationally demanding, and more consistent with the business objective of improving the identification of likely non-completion cases.

5. Time Delay Prediction

5.1 Linear Regression

Linear regression is selected to predict the possible operation duration because the initial correlation analysis showed observable relationships between the target and several variables. This suggests that a linear model could provide a reasonable starting point for prediction. In addition, its interpretability makes it easier to explain the patterns captured by the model and understand how different factors are associated with operation duration.

The correlation analysis suggests that loan-related variables have relatively stronger associations with completion duration. Meanwhile, the average completion time by month indicates that the duration may vary across different periods, suggesting the presence of time-related patterns. In addition, applicant-related characteristics, including income and whether the application is jointly submitted with a second applicant, may also help explain variation in the expected operation duration.

This model is developed using completed applications only, as completion duration is only meaningful when an application has actually completed. Predictor selection is guided by observed relationships between the variables and completion duration. Additional features are also created to better capture loan-related, applicant-related, and time-related patterns.

The model explains around 43% of the variation in completion time, suggesting that loan size, affordability, and applicant financial capacity are key factors influencing expected completion duration. This provides useful business insight for estimating pipeline timing and managing expected lending flows. However, the remaining unexplained variation indicates that some relationships may be non-linear, suggesting that a non-linear model could be useful for capturing more complex completion patterns.

5.2 Non-linear Model for Time Delay Prediction

5.2.1 XGBoost Regression and Results

Linear regression assumes that each variable affects completion time by a fixed amount regardless of context. In practice, this is unlikely to hold — for example, a high LTV loan may behave differently depending on whether the applicant is a joint or single borrower, or whether the property is a new build. XGBoost addresses this by learning from combinations of variables rather than treating each one independently. It is also less sensitive to unusual values in the data, which is useful given that some transactions take significantly longer than average to complete.

The XGBoost model is trained only on completed applications, because completion duration is only observed when an application has actually completed. The model uses a combination of applicant-related, loan-related, property-related, and time-related features, including joint application status, applicant income, property age, LTV, income multiplier, new-build indicator, average monthly delay, and approval timing variables.

To support model validation, the completed training sample is further split into a training subset and a validation subset using an 80:20 chronological split. This allows the model to be evaluated on later observations rather than randomly mixed cases, which is more consistent with the forecasting nature of the business problem.

This model is evaluated on the held-out test set, achieving an R^2 of 0.46, an RMSE of 10.31 days, and an MAE of 8.18 days. Full results are summarised in the comparison table below.

5.2.2 Comparison with linear model

XGBoost performs better than linear regression on all three measures. However, the improvement is relatively small, R^2 increases by around 4 percentage points and the average error falls by roughly half a day. This suggests that while there are some non-linear patterns in the data, they are not the main driver of completion time. The more likely constraint is the information available in the dataset itself, which is discussed in the Limitations section.

XGBoost and linear regression represent different points on the trade-off between predictive accuracy and interpretability. Where a precise forecast is required — for example, in cash flow planning or balance sheet management — XGBoost's marginal improvement in accuracy may be of practical value. Where the priority is communicating results clearly to stakeholders, the linear model's straightforward coefficients may be more appropriate. In practice, both models could serve a role depending on the business context in which they are applied.

Model	R^2	RMSE	MAE
XGBoost	0.4621	10.31 days	8.18 days
Linear Regression	0.4271	10.64 days	8.43 days

Table 2: Comparison of Completion Duration Prediction Models

5.2.3 Interpretation

Despite the moderate overall fit, the model identifies approval timing and affordability-related variables — such as LTV ratio and income multiplier — as the strongest predictors of completion time. This suggests that more financially stretched applicants, and those approved at certain points in the year, tend to experience longer delays. These patterns, while not sufficient to predict individual

completion times with high precision, may still offer useful insight for pipeline management and cash flow planning.

6. Bootstrap Forecast of Two-Month Cashflows

6.1 Objective and selected models

The objective of this section is to produce a probabilistic forecast of mortgage advance cashflows expected to be received within the next two months from cases currently in the open pipeline. Rather than relying on a single-point prediction, a bootstrap simulation framework is adopted to capture model uncertainty and generate a distribution of plausible cashflow outcomes.

Two models developed in earlier sections are selected for this purpose. The Lasso logistic regression model (with regularisation parameter C = 0.0886, selected via cross-validation) is used to estimate the probability that each open case would eventually complete. The XGBoost regression model is used to predict the number of days between approval and advance for each case. Together, these two models determine both whether a case completes and whether it does so within the two-month forecast window.

6.2 Bootstrap design and assumptions

The simulation is run 500 times. In each run, the training data is resampled and both models are retrained, introducing variability that reflects real-world uncertainty. As in model development, the Lasso inputs are standardised using the scaler fitted on the training data, and a fixed random seed is used so that results are reproducible. For each open pipeline case, the model predicts whether it would complete and how long it would take. Only cases completing within 61 days contribute to the cashflow total for that run.

The approach assumes that historical completion patterns remain broadly consistent over the forecast horizon and that open pipeline cases are comparable to historical cases in the training data.

6.3 Estimated cashflow distribution

The 500 simulation runs produce the following cashflow estimates:

Statistic	Value
Mean	£ 87,468,222

Median	£ 87,357,500
5%	£ 82,559,700
95%	£ 92,484,700

Table 3: Simulated Two-Month Cashflow Distribution

The mean and median are closely aligned, indicating a stable and consistent forecast with no significant skew from outlying runs. The 90% confidence interval ranges from £82.6 million to £92.5 million, a spread of approximately £9.9 million, reflecting stable and consistent model predictions.

6.4 Business value for balance-sheet management

This forecast helps the business plan its cash and funding needs more effectively. The central estimate of £87.5 million gives a clear target for two-month cashflow planning, while the 5th percentile of £82.6 million offers a cautious lower bound for worst-case planning. At the other end, the 95th percentile of £92.5 million represents the upper limit under more favourable conditions.

Having visibility of the downside scenario allows lenders to assess in advance whether contingency reserves are needed or whether additional funding should be sourced elsewhere, reducing the risk of unexpected liquidity shortfalls. By using this approach, the business can move away from simple pipeline counts and instead base its funding decisions on a clearer, evidence-based view of expected cashflows, leading to better planning and lower costs.

7. Limitations and Recommendations

The dataset captures applicant-level financial characteristics at the point of approval, but omits key process-level and external variables that are known drivers of both completion probability and time-to-completion. For example, the absence of property chain length, legal processing status, and macroeconomic conditions at approval date means the models are likely to exhibit omitted variable bias. Improving predictive performance may therefore require augmenting the dataset with process-level administrative data, rather than solely refining model architecture.

8. Conclusion

The models developed in this project provide a practical foundation for mortgage pipeline management and cashflow planning. On the business side, the completion probability model can be used to flag high-risk applications early, enabling proactive intervention before cases fall through. The cashflow forecast gives funding teams a more reliable basis for liquidity planning than simple pipeline counts alone.

On the data side, the current models are constrained by the information available at approval. Including process-level variables is likely to deliver more meaningful performance improvements than further model tuning, as the key limitation lies in the data rather than the modelling approach.

Ultimately, the best model is not the most complex one. It is the one that best balances predictive performance, interpretability, and alignment with the decision it is designed to support.

Appendix

Use of Generative AI

Generative AI is used to support the coding and methodological development of this analysis. The main prompts focus on exploratory data analysis and data cleaning steps, printing variables selected by the Lasso model, debugging Python code for the completion probability model, generating Python code to report Accuracy, AUC, and Lasso variable selection, printing R^2, RMSE, and MAE for the time delay prediction model, and applying a temporal train-test split.

Full AI prompt records

1. Please advise me on the steps of EDA and data cleaning for my business analytics modelling report.
2. Please teach me how to print the variables selected from Lasso model.
3. Please help me debug this Python code for the completion probability model.
4. For the completion probability model, I want to see the Accuracy, AUC and Lasso Variable Selection. Please show me the Python code.
5. For the Time Delay Prediction model, please show me how to print R^2, RMSE and MAE.
6. How to do temporal split for my dataset? I want to use earlier records as training data and later records as testing data.